Collaborative Sentiment Analysis Project Report

Team Members : Melek, Amisha, Ariel

Github repository : https://github.com/AmuuLakh/MLOps_Pipeline

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1. Overview of the Chosen Approach

Our project focuses on building an **end-to-end sentiment analysis pipeline using a BERT-based model**, following an **MLOps-oriented approach** to ensure modularity, collaboration, and reproducibility.

The workflow was **inspired by the Kaggle notebook** "Sentiment Analysis using BERT" by Prakhar Rathi.
We adapted its structure and logic but refactored it into independent, testable Python modules, integrated with GitHub, Trello, and continuous integration (CI).

The complete pipeline included:

1. **Data Extraction** – loading and validating raw text data.
2. **Data Cleaning & Normalization** – text preprocessing (lowercasing, removing punctuation, handling emojis, contractions).
3. **Tokenization** – splitting text into tokens using Hugging Face's `AutoTokenizer`.
4. **Model Fine-Tuning** – fine-tuning `bert-base-uncased` on a sentiment dataset.
5. **Inference** – building a function to predict sentiment on new inputs.
6. **Testing & Logging** – automated verification with `pytest` and dedicated log files.

The modular structure followed this layout:

```
project/
├─ requirements.txt
├─ data_cleaning.log
├─ data_log.log
├─ model_training.log
├─ src/
│  ├─ data_extraction.py
│  ├─ data_processing.py
│  ├─ model.py
│  ├─ inference.py
│  └─ data/
│     ├─ processed/
│     │   ├─ eval_tokenized.csv
│     │   └─ train_tokenized.csv
│     └─ dataset.csv
└─ tests/
    └─ unit/
        ├─ test_data_extraction.py
        ├─ test_data_processing.py
        ├─ test_model.py
        └─ test_inference.py
```

This approach allowed each team member to work independently on separate modules while maintaining a coherent development flow through pull requests and reviews.

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2. Division of Labor

Student	Responsibilities
Amisha	Implemented `data_extraction.py`, unit tests, and logging. Assisted with debugging failed tests and documenting the project. Implemented `model_test.py` and `inference_test.py`.
Melek	Collaborated on `data_processing.py` (text cleaning & normalization). Wrote preprocessing tests for the implemented part. Implemented `model.py` to train and save the model. Added the `inference.py` to predict on test text.

Ariel	Created the initial GitHub repository and prepared the project's structure. Collaborated on `data_processing.py` (Tokenization & Dataset Splitting). Wrote preprocessing tests for the implemented part. Worked on the reporting and taking screenshots.

Collaboration was achieved via:

- **GitHub** (branching, pull requests, and code reviews).
- **Trello** (for progress tracking).
- **WhatsApp** (for real-time communication).

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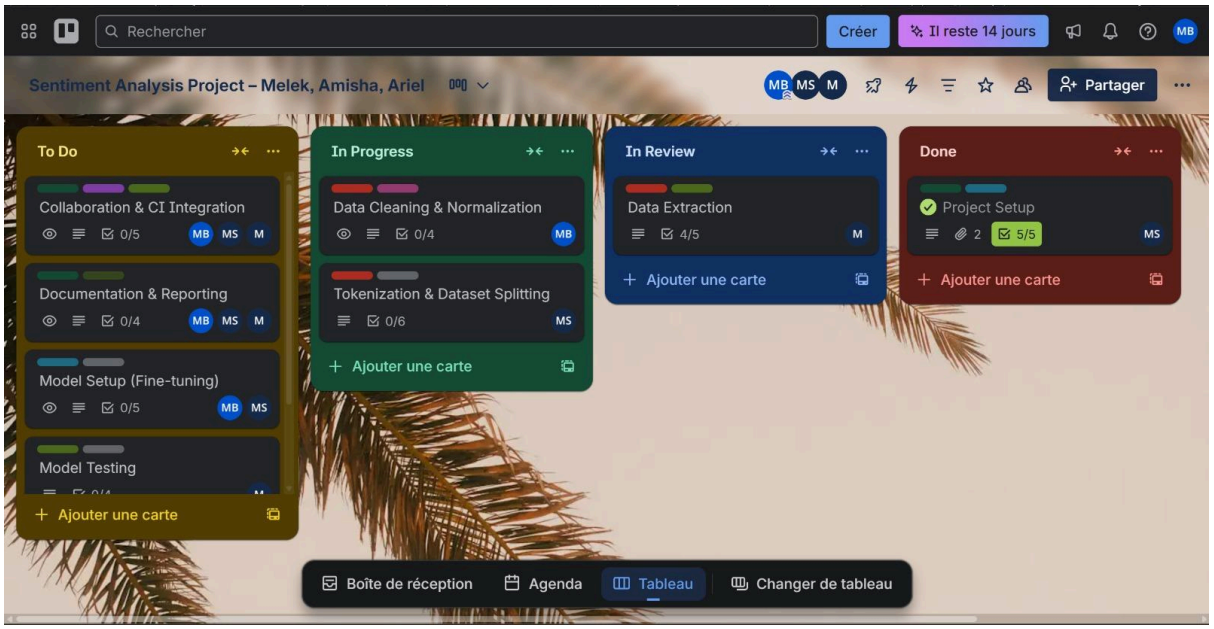
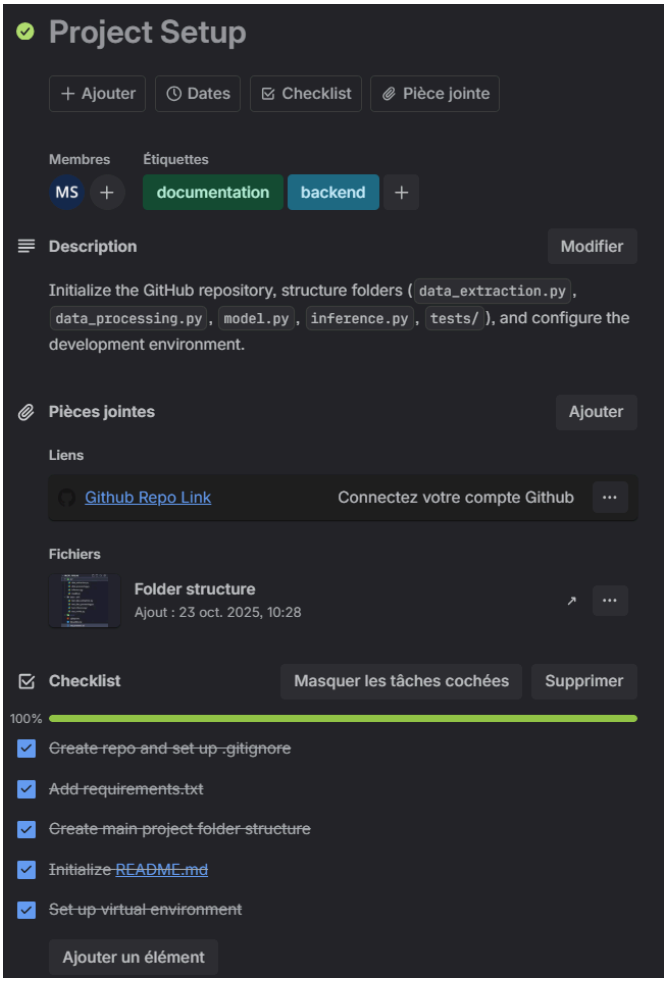
3. Trello Board Screenshots



Board Structure:

- **To Do:** Tasks pending implementation.
- **In Progress:** Ongoing work with assignees.
- **In Review:** Tasks waiting for PR approval.
- **Done:** Completed and verified components.

Example cards:



Each card included:

- **Description** of the task.
- **Checklist** with acceptance criteria.
- **Labels** (e.g., `data`, `model`, `testing`, `documentation`).
- **Attachments** (code snippets, GitHub PR links, test results).

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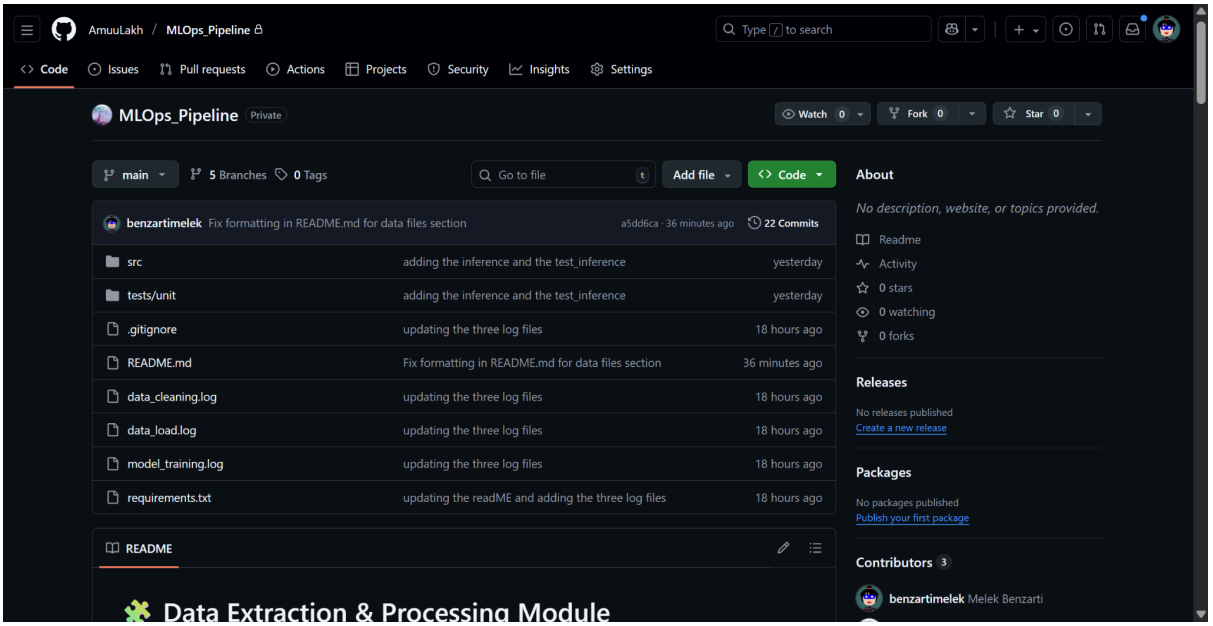
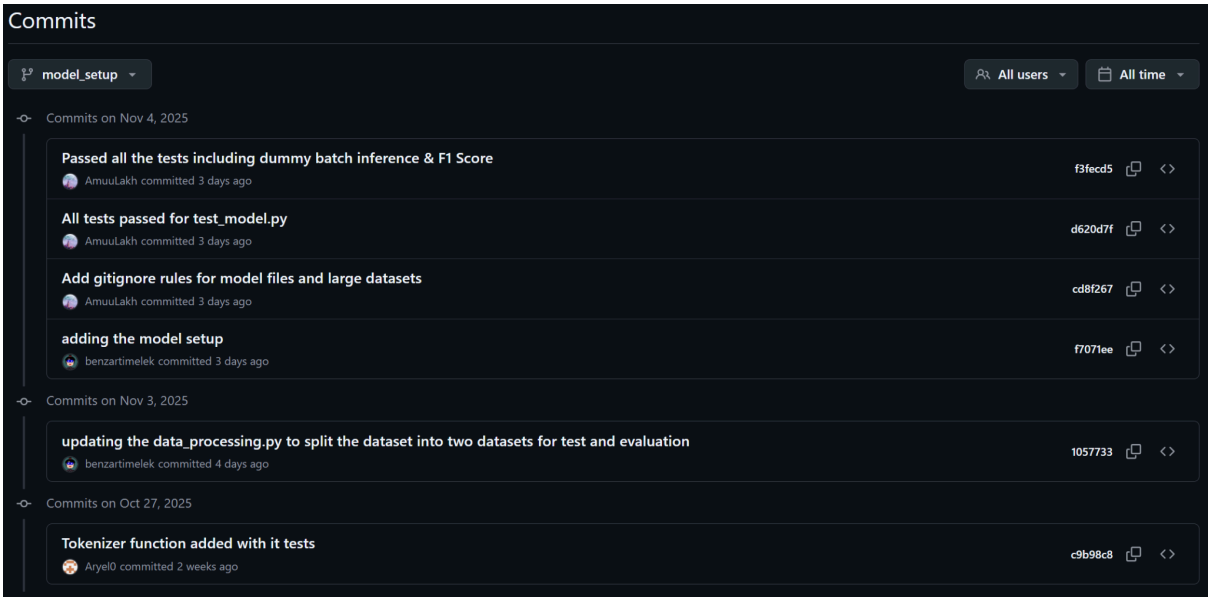
4. GitHub Screenshots of Commits and Pull Requests

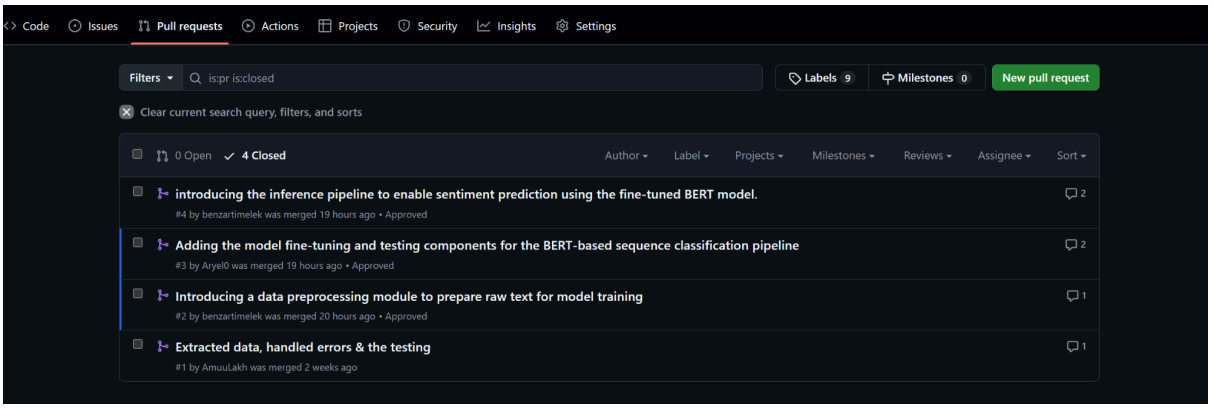
Github Repository (main branch)



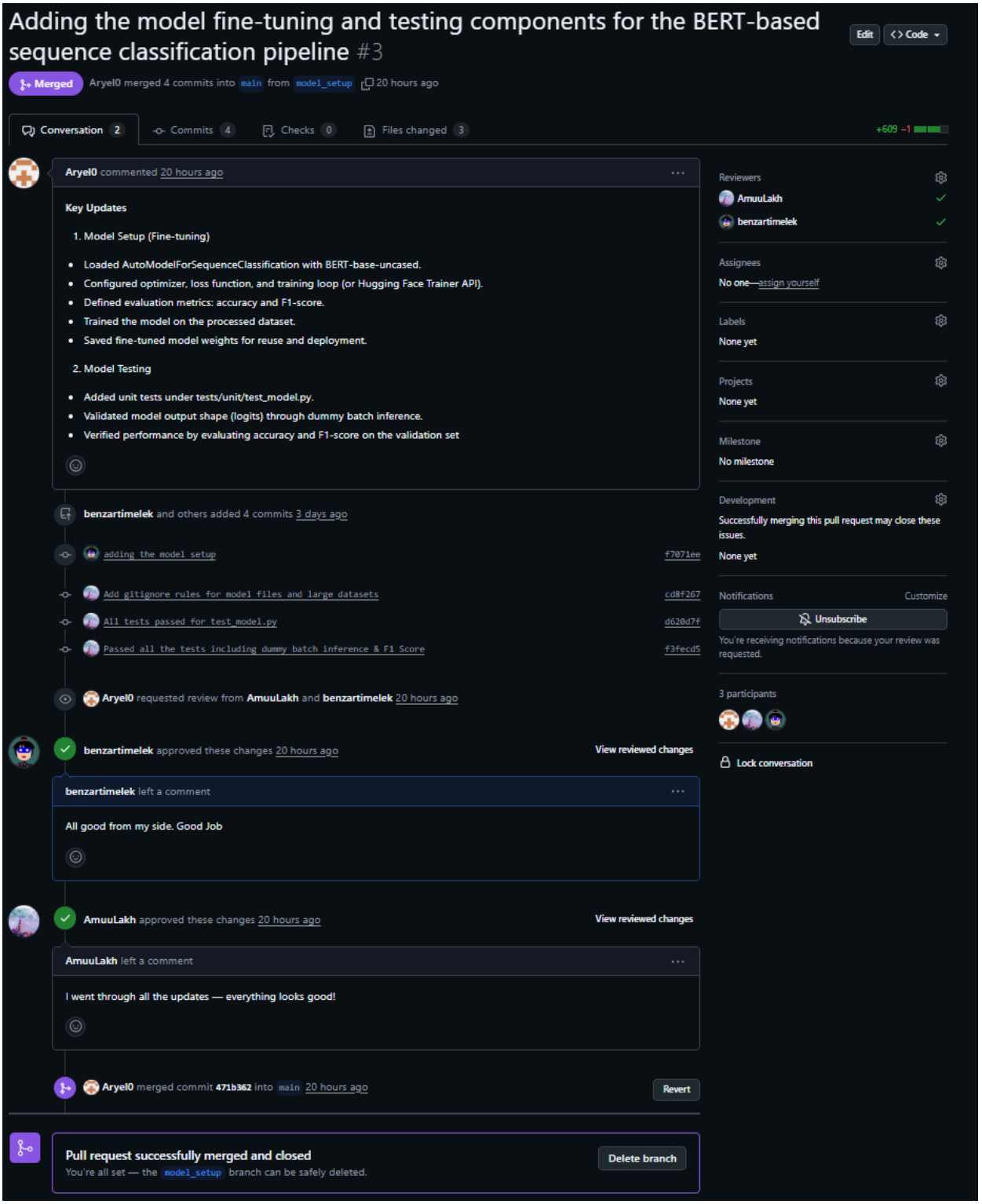
Example of commits to the model_setup branch:



Pull requests:



Example of one pull requests with reviews:



GitHub Collaboration Workflow:

- Each member worked on a **feature branch** (`data_extraction`, `model`, etc.).

- Code was integrated through **pull requests** reviewed by teammates.
- We maintained **clear commit messages**.

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5. Challenges Faced

Despite following a structured pipeline, several practical challenges emerged during the project:

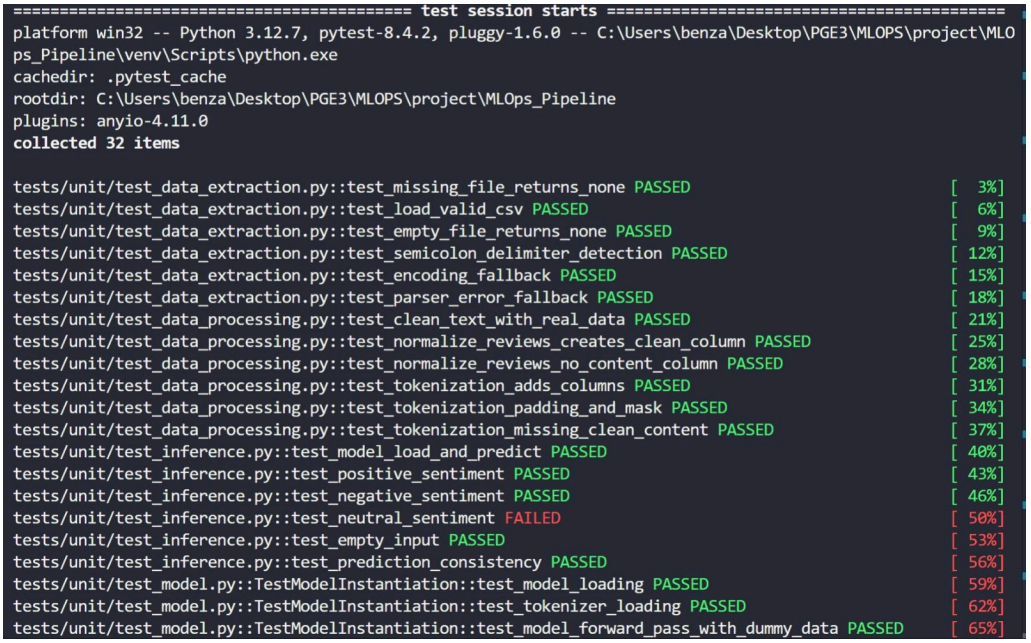
1. Long Training Time

- Fine-tuning the BERT model on our sentiment dataset took **nearly 5 hours**, even after:
    - Reducing the number of epochs.
    - Lowering batch size.
    - Optimizing GPU utilization.

⇒ Switched to a **DistilBERT** model (smaller model).

2. Persistent Test Failure

- One of our inference tests consistently failed.
- After multiple debugging attempts and code adjustments, we discovered that the issue stemmed from an **incompletely trained model**.
- Since retraining would again require 5 hours, we documented this limitation and decided **not to retrain** within the current project timeline.



```
tests/unit/test_data_extraction.py::test_encoding_fallback PASSED                                [ 15%]
tests/unit/test_data_extraction.py::test_parser_error_fallback PASSED                            [ 18%]
tests/unit/test_data_processing.py::test_clean_text_with_real_data PASSED                        [ 21%]
tests/unit/test_data_processing.py::test_normalize_reviews_creates_clean_column PASSED           [ 25%]
tests/unit/test_data_processing.py::test_normalize_reviews_no_content_column PASSED              [ 28%]
tests/unit/test_data_processing.py::test_tokenization_adds_columns PASSED                        [ 31%]
tests/unit/test_data_processing.py::test_tokenization_padding_and_mask PASSED                    [ 34%]
tests/unit/test_data_processing.py::test_tokenization_missing_clean_content PASSED               [ 37%]
tests/unit/test_inference.py::test_model_load_and_predict PASSED                                 [ 40%]
tests/unit/test_inference.py::test_positive_sentiment PASSED                                     [ 43%]
tests/unit/test_inference.py::test_negative_sentiment PASSED                                     [ 46%]
tests/unit/test_inference.py::test_neutral_sentiment FAILED                                      [ 50%]
tests/unit/test_inference.py::test_empty_input PASSED                                            [ 53%]
tests/unit/test_inference.py::test_prediction_consistency PASSED                                 [ 56%]
tests/unit/test_model.py::TestModelInstantiation::test_model_loading PASSED                      [ 59%]
tests/unit/test_model.py::TestModelInstantiation::test_tokenizer_loading PASSED                  [ 62%]
tests/unit/test_model.py::TestModelInstantiation::test_model_forward_pass_with_dummy_data PASSED [ 65%]
tests/unit/test_model.py::TestModelInstantiation::test_model_output_logits_range PASSED          [ 68%]
tests/unit/test_model.py::TestModelInstantiation::test_model_predictions_sum_to_one PASSED       [ 71%]
tests/unit/test_model.py::TestComputeMetrics::test_compute_metrics_perfect_predictions PASSED    [ 75%]
tests/unit/test_model.py::TestComputeMetrics::test_compute_metrics_all_wrong PASSED              [ 78%]
tests/unit/test_model.py::TestComputeMetrics::test_compute_metrics_returns_correct_keys PASSED   [ 81%]
tests/unit/test_model.py::TestPrepareDatasets::test_prepare_datasets_structure PASSED            [ 84%]
tests/unit/test_model.py::TestModelWithRealTokenizer::test_model_with_tokenized_text PASSED      [ 87%]
tests/unit/test_model.py::TestDummyBatchInference::test_run_dummy_batch_inference PASSED         [ 90%]
tests/unit/test_model.py::TestDummyBatchInference::test_compare_predictions_to_expected_shape PASSED [ 93%]
tests/unit/test_model.py::TestDummyBatchInference::test_evaluate_accuracy_and_f1_score PASSED    [ 96%]
tests/unit/test_model.py::TestDummyBatchInference::test_batch_inference_consistency PASSED       [100%]
```

3. Dependency & Version Issues

- Several library mismatches (especially with `transformers` and `torch`) caused runtime errors on different machines.

- This was solved by syncing the `requirements.txt` and enforcing a consistent environment using Python 3.12.

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6. Future Improvements

If given more time, the following enhancements would further improve the project:

1. **Train with a Larger Dataset**
   Improve generalization and accuracy by using a more diverse, large-scale dataset (e.g., IMDb or Yelp reviews).

2. **Reduce Training Time**
   - Use smaller, optimized transformer architectures.

   - Explore gradient accumulation or mixed-precision training.

3. **Fix the Failed Test with Retraining**
   - Fully retrain the model for all epochs to ensure inference outputs match expected shapes.

   - Validate using a clean test split.

4. **Improve Deployment**
    - Wrap the inference script into a **Flask or FastAPI** endpoint for real-time predictions.
    - Add Docker support for reproducibility.
5. **Enhanced CI/CD Pipeline**
   Automate model retraining and deployment using **GitHub Actions** or **MLflow**.
6. **User Interface**
   Create a minimal web app allowing users to type text and get live sentiment predictions.

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7. Conclusion

This project successfully demonstrates the principles of **MLOps**, teamwork, and reproducible AI development.
Despite facing runtime and training limitations, our modular structure, testing pipeline, and documentation reflect solid engineering practices.

Each member contributed effectively within their role, ensuring that the resulting pipeline — even with its constraints — remains functional, transparent, and extensible for future improvements.